Sea Star Microbiomes Sequence Processing

Tags	16S-V4 sea star
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Description	Sequence Processing code compiled and run by Colleen Kellogg

Code used for sequence QC/QC for ***Multi-year study of sea star microbiomes reveals regional and host-specific differences and differentially abundant taxa across wasting disease stages*** dataset

Raw sequence data is processed using QIIME2. The V4 region of the 16S rRNA gene was amplified using fusion primers (515F-Y / 806RB) based on template from André Comeau from the Microbiome Helper paper.

A total of 8 sequencing runs were done for this sample set. Each run was processed using the same settings to facilitate merging the datasets.

Run 1

Run 2

Run 3

Runs 4 - 8

Run 1

Activate QIIME2

```
source activate qiime2-2022.8
```

Make working directory

```
 mkdir 16SV4_seastar_mussels_sept2022
```

Create symbolic links (symlink) for all files in Fastq directory on storage server in my directory on the analysis server.

```
ln -s /mnt/MiSeq/16S_SeaStar_Mussels_Run20220901/Run20220901/Alignment_1/20220904_003035/
Fastq/*.fastq.gz 16SV4_seastar_mussels_sept2022/reads
```

Import reads into QIIME2

```
nohup qiime tools import \
   --type SampleData[PairedEndSequencesWithQuality] \
   --input-path /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/reads \
   --output-path /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/reads_qza/smic_run1
_sept2022.qza \
   --input-format CasavaOneEightSingleLanePerSampleDirFmt &
```

Make directory for processed data

```
mkdir /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022
```

Run specs before trimming

```
nohup qiime demux summarize \
   --i-data /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/reads_qza/smic_run1_sept
2022.qza \
   --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/smic_run1_sept20
22.qzv &
```

Trim primers

```
nohup qiime cutadapt trim-paired \
   --i-demultiplexed-sequences /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/reads
_qza/smic_run1_sept2022.qza \
   --p-cores 2 \
   --p-front-f GTGYCAGCMGCCGCGGTAA \
   --p-front-r GGACTACNVGGGTWTCTAAT \
```

```
   --p-discard-untrimmed \
   --p-no-indels \
   --o-trimmed-sequences /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/reads_qza/s
mic_run1_sept2022_trimmed.qza &
```

Run specs after trimming primers

```
nohup qiime demux summarize \
   --i-data /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/reads_qza/smic_run1_sept
2022_trimmed.qza \
   --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/smic_run1_sept20
22_trimmed.qzv &
```

Review read quality summary smic_run1_sept2022_trimmed.qzv **in** QIIME2 view

Run dada2 (based on how the read quality looks)

```
nohup qiime dada2 denoise-paired \
    --i-demultiplexed-seqs /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/reads_qz
a/smic_run1_sept2022_trimmed.qza \
    --p-trunc-len-f 230 \
    --p-trunc-len-r 190 \
    --p-max-ee-f 4 \
    --p-max-ee-r 6 \
    --p-n-threads 8 \
    --output-dir /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output &
```

Export dada2 stats and review

```
qiime tools export --input-path /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada
2_output/denoising_stats.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_run1_sep
t2022/dada2_output/
```

Check out the representative sequences and ASV table in QIIME2 view

```
qiime feature-table tabulate-seqs \
  --i-data /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/representati
ve_sequences.qza \
  --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/dada2_output/rep-
seqs.qzv
```

```
qiime feature-table summarize \
  --i-table /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/table.qza \
  --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/dada2_output/tabl
e_summary.qzv \
  --m-sample-metadata-file /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/smic-run1-
sept2022-metadata.tsv
```

Filter out rare ASVs [do this per run, before merging]

From Microbiome Helper: *Based on the above summary visualization you can choose a cut-off for how frequent a variant needs to be (and optionally how many samples need to have the variant) for it to be retained. One possible choice would be to remove all ASVs that have a frequency of less than 0.1% of the mean sample depth. This cut-off excludes ASVs that are likely due to MiSeq bleed-through between runs (reported by Illumina to be 0.1% of reads). To calculate this cut-off you would identify the mean sample depth in the above visualization, multiply it by 0.001, and round to the nearest integer.*

Across the whole sample set, the average frequency per sample is 25,831. But if you remove samples with reads less than 6000, the average sampling depth is 31945. 0.1% of the latter is 31 and the former is 25.

```
qiime feature-table filter-features --i-table /home/colleen.kellogg/16SV4_seastar_mussels
_sept2022/dada2_output/table.qza \
    --p-min-frequency 26 \
    --p-min-samples 2 \
    --o-filtered-table /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/
table_filt.qza
```

Remove ASVs that were filtered out

from representative sequences

```
qiime feature-table filter-seqs --i-data /home/colleen.kellogg/16SV4_seastar_mussels_sept
2022/dada2_output/representative_sequences.qza \
    --i-table /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/table_fil
t.qza \
    --o-filtered-data /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/r
epseqs_filt.qza
```

Summarize filtered feature table

```
qiime feature-table summarize \
  --i-table   /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/table_fil
t.qza \
  --o-visualization  /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/fi
lt_table_summary.qzv \
  --m-sample-metadata-file /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/smic-run1-
sept2022-metadata.tsv
```

Classify reads

```
nohup qiime feature-classifier classify-sklearn \
    --i-reads /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/repseqs_f
ilt.qza \
    --i-classifier /home/colleen.kellogg/dbs_qiime2022/silva-138-99-515-806-nb-classifie
r.qza \
    --p-n-jobs 8 \
    --output-dir /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/taxa_sklearn &
```

Filter chloroplasts and mitochondria from feature table

```
qiime taxa filter-table \
  --i-table /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/table_filt.
qza \
  --i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/taxa_sklearn/classific
ation.qza \
  --p-exclude mitochondria,chloroplast,eukaryota \
  --o-filtered-table /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/ta
ble_filt_NCM.qza
```

Filter chloroplasts and mitochondria from rep seqs

```
qiime taxa filter-seqs --i-sequences /home/colleen.kellogg/16SV4_seastar_mussels_sept202
2/dada2_output/repseqs_filt.qza \
  --i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/taxa_sklearn/classific
ation.qza \
  --p-exclude mitochondria,chloroplast,eukaryota \
  --o-filtered-sequences /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_outpu
t/repseqs_filt_NCM.qza
```

Export files for analysis R

ASV table

```
qiime tools export --input-path /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/table_filt_NCM.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/exported/
```

Representative Sequences

```
qiime tools export --input-path /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/repseqs_filt_NCM.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/exported/
```

Taxonomy

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/taxa_sklearn/classification.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/exported/
```

Make BIOM file

Join the taxonomy data with the feature table.

Don’t forget to change the taxonomy header so that it is as follows:

#OTUID taxonomy confidence

```
biom add-metadata -i /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/exported/feature-table.biom -o /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/exported/feature-table-tax.biom --observation-metadata-fp /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/exported/taxonomy.tsv --sc-separated taxonomy
```

Convert BIOM file to text file

```
biom convert -i /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/exported/feature-table-tax.biom -o /mnt/Processed/ckellogg/sea-star/smic_run1_sept2022/exported/smic_run1_ncmtable_tax.tsv --to-tsv --header-key taxonomy
```

Run 2

Fire up QIIME2

```
source activate qiime2-2022.8
```

Make working directory

```
 mkdir 16SV4_seastar_run2_nov2022
```

Create symbolic links (symlink) for all files in Fastq directory on storage server in my directory on the analysis server.

```
ln -s /mnt/MiSeq/16S_SSWD_Run20221105/Run20221105/Alignment_2/20221108_091551/Fastq/*.fas
tq.gz /home/colleen.kellogg/16SV4_seastar_run2_nov2022/reads
```

Import reads into QIIME2

```
nohup qiime tools import \
   --type SampleData[PairedEndSequencesWithQuality] \
   --input-path /home/colleen.kellogg/16SV4_seastar_run2_nov2022/reads \
   --output-path /home/colleen.kellogg/16SV4_seastar_run2_nov2022/reads_qza/smic_run2_nov
2022.qza \
   --input-format CasavaOneEightSingleLanePerSampleDirFmt &
```

Make directory for processed data

```
mkdir /mnt/Processed/ckellogg/sea-star/smic_run2_nov2022
```

Run specs before trimming

```
nohup qiime demux summarize \
   --i-data /home/colleen.kellogg/16SV4_seastar_run2_nov2022/reads_qza/smic_run2_nov2022.
qza \
```

```
   --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run2_nov2022/smic_run2_nov202
2.qzv &
```

Trim primers

```
nohup qiime cutadapt trim-paired \
   --i-demultiplexed-sequences /home/colleen.kellogg/16SV4_seastar_run2_nov2022/reads_qz
a/smic_run2_nov2022.qza \
   --p-cores 4 \
   --p-front-f GTGYCAGCMGCCGCGGTAA \
   --p-front-r GGACTACNVGGGTWTCTAAT \
   --p-discard-untrimmed \
   --p-no-indels \
   --o-trimmed-sequences /home/colleen.kellogg/16SV4_seastar_run2_nov2022/reads_qza/smic_
run2_nov2022_trimmed.qza &
```

Run specs after trimming

```
nohup qiime demux summarize \
   --i-data /home/colleen.kellogg/16SV4_seastar_run2_nov2022/reads_qza/smic_run2_nov2022_
trimmed.qza \
   --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run2_nov2022/smic_run2_nov2022
_trimmed.qzv &
```

Review read quality summary smic_run2_nov2022_trimmed.qzv **in** QIIME2 view

Run dada2 (based on how the read quality looks)

```
nohup qiime dada2 denoise-paired \
    --i-demultiplexed-seqs /home/colleen.kellogg/16SV4_seastar_run2_nov2022/reads_qza/smi
c_run2_nov2022_trimmed.qza \
    --p-trunc-len-f 230 \
    --p-trunc-len-r 190 \
    --p-max-ee-f 4 \
    --p-max-ee-r 6 \
    --p-n-threads 4 \
    --output-dir /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output &
```

Export dada2 stats and review

```
qiime tools export --input-path /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_ou
tput/denoising_stats.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_run2_nov202
```

```
2/dada2_output/
```

Stats summary

ave % retained	ave number of reads
75.7633333	62239.6866

```
qiime feature-table tabulate-seqs \
  --i-data /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/representative_s
equences.qza \
  --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run2_nov2022/dada2_output/rep-s
eqs-smic-run2.qzv
```

```
qiime feature-table summarize \
  --i-table /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/table.qza \
  --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run2_nov2022/dada2_output/table
_summary.qzv \
  --m-sample-metadata-file /mnt/Processed/ckellogg/sea-star/smic_run2_nov2022/smic-run2-n
ov2022-metadata.tsv
```

Filter out rare ASVs -do this per run, before merging

Across the whole sample set, the average frequency per sample is 57,163. But if you remove samples with reads less than 6000, the average sampling depth is 61,951. 0.1% of the latter is 61 and the former is 57.

```
qiime feature-table filter-features --i-table /home/colleen.kellogg/16SV4_seastar_run2_no
v2022/dada2_output/table.qza \
    --p-min-frequency 57 \
    --p-min-samples 2 \
    --o-filtered-table /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/tabl
e_filt_smic_run2.qza
```

Remove ASVs that were filtered out

From representative sequences

```
qiime feature-table filter-seqs --i-data /home/colleen.kellogg/16SV4_seastar_run2_nov202
2/dada2_output/representative_sequences.qza \
    --i-table /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/table_filt_sm
```

```
ic_run2.qza \
    --o-filtered-data /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/repse
qs_filt.qza
```

Summarize filtered feature table

```
qiime feature-table summarize \
  --i-table   /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/table_filt_sm
ic_run2.qza \
  --o-visualization  /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/filt_t
able_summary_smic_run2.qzv \
  --m-sample-metadata-file /mnt/Processed/ckellogg/sea-star/smic_run2_nov2022/smic-run2-n
ov2022-metadata.tsv
```

Merge Run 1 and Run 2

Make a directory for merged data

```
mkdir /home/colleen.kellogg/16SV4_smic_merged/
```

Merge feature table

```
qiime feature-table merge \
--i-tables /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/table_filt.q
za \
--i-tables /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/table_filt_smic_
run2.qza \
--o-merged-table /home/colleen.kellogg/16SV4_smic_merged/smic_runs12_mergedtable.qza
```

Merge representative sequences

```
qiime feature-table merge-seqs \
--i-data /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/repseqs_filt.q
za \
--i-data /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/repseqs_filt.qza \
--o-merged-data /home/colleen.kellogg/16SV4_smic_merged/smic_runs12_merged_repseqs.qza
```

Summarize Merged Feature Table and Rep Seqs

```
qiime feature-table summarize \
--i-table /home/colleen.kellogg/16SV4_smic_merged/smic_runs12_mergedtable.qza \
--o-visualization /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12_mergedtab
le_summary.qzv \
--m-sample-metadata-file /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_merged_me
tadata.tsv
```

```
qiime feature-table tabulate-seqs \
  --i-data /home/colleen.kellogg/16SV4_smic_merged/smic_runs12_merged_repseqs.qza \
  --o-visualization /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12_merged_
repseqs_summary.qzv
```

Classify reads

```
nohup qiime feature-classifier classify-sklearn \
    --i-reads /home/colleen.kellogg/16SV4_smic_merged/smic_runs12_merged_repseqs.qza \
    --i-classifier /home/colleen.kellogg/dbs_qiime2022/silva-138-99-515-806-nb-classifie
r.qza \
    --p-n-jobs 10 \
    --output-dir /mnt/Processed/ckellogg/sea-star/smic_merged_runs/taxa_sklearn &
```

Filter chloroplasts and mitochondria from from feature table

```
qiime taxa filter-table \
  --i-table /home/colleen.kellogg/16SV4_smic_merged/smic_runs12_mergedtable.qza \
  --i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_merged_runs/taxa_sklearn/classificat
ion.qza \
  --p-exclude mitochondria,chloroplast,eukaryota \
  --o-filtered-table /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12_merged
table_NCM.qza
```

Filter chloroplasts and mitochondria from Rep Seqs

```
qiime taxa filter-seqs --i-sequences /home/colleen.kellogg/16SV4_smic_merged/smic_runs12_
merged_repseqs.qza \
  --i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_merged_runs/taxa_sklearn/classificat
ion.qza  \
  --p-exclude mitochondria,chloroplast,eukaryota \
```

```
  --o-filtered-sequences /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12_me
rged_repseqs_NCM.qza
```

Export files for analysis in R

ASV table

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_ru
ns12_mergedtable_NCM.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/
exported/
```

Representative sequences

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_ru
ns12_merged_repseqs_NCM.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_merged_ru
ns/exported/
```

Taxonomy

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/taxa_sk
learn/classification.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/
exported/
```

Make BIOM file

Change the taxonomy header so that it is as follows:

#OTUID taxonomy confidence

```
biom add-metadata -i /mnt/Processed/ckellogg/sea-star/smic_merged_runs/exported/feature-t
able.biom -o /mnt/Processed/ckellogg/sea-star/smic_merged_runs/exported/feature-table-ta
x.biom --observation-metadata-fp /mnt/Processed/ckellogg/sea-star/smic_merged_runs/export
ed/taxonomy.tsv --sc-separated taxonomy
```

```
biom convert -i /mnt/Processed/ckellogg/sea-star/smic_merged_runs/exported/feature-table-
tax.biom -o /mnt/Processed/ckellogg/sea-star/smic_merged_runs/exported/smic_run12_merged_
ncmtable_tax.tsv --to-tsv --header-key taxonomy
```

Run 3

Fire up QIIME2

```
source activate qiime2-2022.8
export TMPDIR='/home/colleen.kellogg/qiimetmp/' #specify tmp drive location
```

Make a working directory

```
 mkdir 16SV4_seastar_run3_april2023
```

Create symbolic links (symlink) for all files in Fastq directory on storage server in my directory on the analysis server.

```
ln -s /mnt/MiSeq/SSWD_Library3_Run20230412/Run20230412/Alignment_1/20230414_154930/Fastq/
*.fastq.gz /home/colleen.kellogg/16SV4_seastar_run3_april2023/reads
```

Import reads into QIIME2

```
nohup qiime tools import \
   --type SampleData[PairedEndSequencesWithQuality] \
   --input-path /home/colleen.kellogg/16SV4_seastar_run3_april2023/reads \
   --output-path /home/colleen.kellogg/16SV4_seastar_run3_april2023/reads_qza/smic_run3_a
pr2023.qza \
   --input-format CasavaOneEightSingleLanePerSampleDirFmt &
```

Make a directory for processed data

```
mkdir /mnt/Processed/ckellogg/sea-star/smic_run3_apr2023
```

Run specs before trimming

```
nohup qiime demux summarize \
   --i-data /home/colleen.kellogg/16SV4_seastar_run3_april2023/reads_qza/smic_run3_apr202
3.qza \
```

```
   --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run3_apr2023/smic_run3_apr202
3.qzv &
```

Trim Primers

```
nohup qiime cutadapt trim-paired \
   --i-demultiplexed-sequences /home/colleen.kellogg/16SV4_seastar_run3_april2023/reads_q
za/smic_run3_apr2023.qza \
   --p-cores 6 \
   --p-front-f GTGYCAGCMGCCGCGGTAA \
   --p-front-r GGACTACNVGGGTWTCTAAT \
   --p-discard-untrimmed \
   --p-no-indels \
   --o-trimmed-sequences /home/colleen.kellogg/16SV4_seastar_run3_april2023/reads_qza/smi
c_run3_apr2023_trimmed.qza &
```

Run specs after trimming

```
nohup qiime demux summarize \
    --i-data /home/colleen.kellogg/16SV4_seastar_run3_april2023/reads_qza/smic_run3_apr20
23_trimmed.qza \
   --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run3_apr2023/smic_run3_apr2023
_trimmed.qzv &
```

Review read quality summary smic_run3_apr2023_trimmed.qzv **in** QIIME2 view

Run dada2 (based on how the read quality looks)

```
nohup qiime dada2 denoise-paired \
    --i-demultiplexed-seqs /home/colleen.kellogg/16SV4_seastar_run3_april2023/reads_qza/s
mic_run3_apr2023_trimmed.qza \
    --p-trunc-len-f 230 \
    --p-trunc-len-r 190 \
    --p-max-ee-f 4 \
    --p-max-ee-r 6 \
    --p-n-threads 8 \
    --output-dir /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output &
```

Export dada2 stats and review

```
qiime tools export --input-path /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_
output/denoising_stats.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_run3_apr20
```

```
23/dada2_output/
```

Check out the representative sequences and ASV table in QIIME2 view

```
qiime feature-table tabulate-seqs \
  --i-data /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/representative
_sequences.qza \
  --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run3_apr2023/dada2_output/rep-s
eqs-smic-run3.qzv
```

```
qiime feature-table summarize \
  --i-table /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/table.qza \
  --o-visualization /mnt/Processed/ckellogg/sea-star/smic_run3_apr2023/dada2_output/table
_summary.qzv \
  --m-sample-metadata-file /mnt/Processed/ckellogg/sea-star/smic_run3_apr2023/smic-run3-a
pr2023-metadata.tsv
```

Filter out rare ASVs -do this per run, before merging

Across the whole sample set, the average frequency per sample is 37385.967. 0.1% is 37.

```
qiime feature-table filter-features --i-table /home/colleen.kellogg/16SV4_seastar_run3_ap
ril2023/dada2_output/table.qza \
    --p-min-frequency 37 \
    --p-min-samples 2 \
    --o-filtered-table /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/ta
ble_filt_smic_run3.qza
```

Remove ASVs that were filtered out.

```
qiime feature-table filter-seqs --i-data /home/colleen.kellogg/16SV4_seastar_run3_april20
23/dada2_output/representative_sequences.qza \
    --i-table /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/table_filt_
smic_run3.qza \
    --o-filtered-data /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/rep
seqs_filt.qza
```

Summarize filtered table

```
qiime feature-table summarize \
  --i-table   /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/table_filt_
smic_run3.qza \
  --o-visualization  /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/filt
_table_summary_smic_run3.qzv \
  --m-sample-metadata-file /mnt/Processed/ckellogg/sea-star/smic_run3_apr2023/smic-run3-a
pr2023-metadata.tsv
```

Merge Run 1, Run 2, and Run 3 ASV tables

```
qiime feature-table merge \
--i-tables /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/table_filt.q
za \
--i-tables /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/table_filt_smic_
run2.qza \
--i-tables /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/table_filt_smi
c_run3.qza \
--o-merged-table /home/colleen.kellogg/16SV4_smic_merged/smic_runs123_mergedtable.qza
```

Merge Run 1, Run 2, and Run 3 Representative Sequences

```
qiime feature-table merge-seqs \
--i-data /home/colleen.kellogg/16SV4_seastar_mussels_sept2022/dada2_output/repseqs_filt.q
za \
--i-data /home/colleen.kellogg/16SV4_seastar_run2_nov2022/dada2_output/repseqs_filt.qza \
--i-data /home/colleen.kellogg/16SV4_seastar_run3_april2023/dada2_output/repseqs_filt.qza
\
--o-merged-data /home/colleen.kellogg/16SV4_smic_merged/smic_runs123_merged_repseqs.qza
```

Summarize Merged Feature Table and Rep Seqs

```
qiime feature-table summarize \
--i-table /home/colleen.kellogg/16SV4_smic_merged/smic_runs123_mergedtable.qza \
--o-visualization /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs123_mergedta
ble_summary.qzv \
--m-sample-metadata-file /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_merged_me
tadata.tsv
```

```
qiime feature-table tabulate-seqs \
  --i-data /home/colleen.kellogg/16SV4_smic_merged/smic_runs123_merged_repseqs.qza \
```

```
  --o-visualization /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs123_merged
_repseqs_summary.qzv
```

Classify reads

```
nohup qiime feature-classifier classify-sklearn \
    --i-reads /home/colleen.kellogg/16SV4_smic_merged/smic_runs123_merged_repseqs.qza \
    --i-classifier /home/colleen.kellogg/dbs_qiime2022/silva/silva-138-99-515-806-nb-clas
sifier.qza \
    --p-n-jobs 10 \
    --output-dir /mnt/Processed/ckellogg/sea-star/smic_merged_runs/taxa_sklearn &
```

Filter chloroplasts and mitochondria from feature table

```
qiime taxa filter-table \
  --i-table /home/colleen.kellogg/16SV4_smic_merged/smic_runs123_mergedtable.qza \
  --i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_merged_runs/taxa_sklearn/classificat
ion.qza \
  --p-exclude mitochondria,chloroplast,eukaryota \
  --o-filtered-table /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs123_merge
dtable_NCM.qza
```

Filter chloroplasts and mitochondria from Rep Seqs

```
qiime taxa filter-seqs --i-sequences /home/colleen.kellogg/16SV4_smic_merged/smic_runs123
_merged_repseqs.qza \
  --i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_merged_runs/taxa_sklearn/classificat
ion.qza  \
  --p-exclude mitochondria,chloroplast,eukaryota \
  --o-filtered-sequences /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs123_m
erged_repseqs_NCM.qza
```

Export Files

ASV Table

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_ru
ns123_mergedtable_NCM.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_merged_run
s/exported/
```

Representative sequences

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_ru
ns123_merged_repseqs_NCM.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_merged_r
uns/exported/
```

Taxonomy

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/taxa_sk
learn/classification.qza --output-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/
exported/
```

Make BIOM file

Change the taxonomy header so that it is as follows:

#OTUID taxonomy confidence

```
biom add-metadata -i /mnt/Processed/ckellogg/sea-star/smic_merged_runs/exported/feature-t
able.biom -o /mnt/Processed/ckellogg/sea-star/smic_merged_runs/exported/feature-table-ta
x.biom --observation-metadata-fp /mnt/Processed/ckellogg/sea-star/smic_merged_runs/export
ed/taxonomy.tsv --sc-separated taxonomy
```

```
biom convert -i /mnt/Processed/ckellogg/sea-star/smic_merged_runs/exported/feature-table-
tax.biom -o /mnt/Processed/ckellogg/sea-star/smic_merged_runs/exported/smic_run123_merged
_ncmtable_tax.tsv --to-tsv --header-key taxonomy
```

Runs 4 - 8

Make Working Directories for Each Run

```
#SSWD_Run4_Run20231013

/mnt/Genomics/MiSeq/SSWD_Run4_Run20231013/Run20231013/Alignment_1/20231016_020219/Fastq

 #SSWD_Run5_Run20240315
```

```
/mnt/Genomics/MiSeq/SSWD_Run5_Run20240315/Run20240315/Alignment_1/20240318_054700/Fastq

#16S_SSWD_Run6_Run20240509

/mnt/Genomics/MiSeq/16S_SSWD_Run6_Run20240509/Run20240509/Alignment_1/20240512_040837/Fas
tq

#16S_SSWD_Run7_Run20240514

/mnt/Genomics/MiSeq/16S_SSWD_Run7_Run20240514/Run20240514/Alignment_1/20240517_035253/Fas
tq

#16S_SSWD_Run8_20240607

/mnt/Genomics/MiSeq/16S_SSWD_Run8_20240607/Run20240607/Alignment_1/20240610_021123/Fastq
```

Make directories for processed data

```
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads
```

Create symbolic links (symlink) for all files in Fastq directory on storage server in my directory on the analysis server.

```
ln -s /mnt/Genomics/MiSeq/SSWD_Run4_Run20231013/Run20231013/Alignment_1/20231016_020219/F
astq/*.fastq.gz /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads

ln -s /mnt/Genomics/MiSeq/SSWD_Run5_Run20240315/Run20240315/Alignment_1/20240318_054700/F
astq/*.fastq.gz /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads

ln -s /mnt/Genomics/MiSeq/16S_SSWD_Run6_Run20240509/Run20240509/Alignment_1/20240512_0408
37/Fastq/*.fastq.gz /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads

ln -s /mnt/Genomics/MiSeq/16S_SSWD_Run7_Run20240514/Run20240514/Alignment_1/20240517_0352
53/Fastq/*.fastq.gz /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads

ln -s /mnt/Genomics/MiSeq/16S_SSWD_Run8_20240607/Run20240607/Alignment_1/20240610_021123/
Fastq/*.fastq.gz /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads
```

Make directories for imported reads

```
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads_qza/
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads_qza/
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads_qza/
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads_qza/
mkdir /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads_qza/
```

Import reads into QIIME2

```
nohup qiime tools import \
   --type SampleData[PairedEndSequencesWithQuality] \
   --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads \
   --output-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads_qza/smic_
run4_oct2023.qza \
   --input-format CasavaOneEightSingleLanePerSampleDirFmt &

nohup qiime tools import \
   --type SampleData[PairedEndSequencesWithQuality] \
   --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads \
   --output-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads_qza/smic_
run5_apr2024.qza \
   --input-format CasavaOneEightSingleLanePerSampleDirFmt &

nohup qiime tools import \
   --type SampleData[PairedEndSequencesWithQuality] \
   --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads \
   --output-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads_qza/smic_
run6_may2024.qza \
   --input-format CasavaOneEightSingleLanePerSampleDirFmt &

nohup qiime tools import \
   --type SampleData[PairedEndSequencesWithQuality] \
   --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads \
   --output-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads_qza/smic_
run7_may2024.qza \
   --input-format CasavaOneEightSingleLanePerSampleDirFmt &

nohup qiime tools import \
   --type SampleData[PairedEndSequencesWithQuality] \
   --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads \
   --output-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads_qza/smic_
run8_june2024.qza \
   --input-format CasavaOneEightSingleLanePerSampleDirFmt &
```

Review run specs before trimming

```
nohup qiime demux summarize \
  --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads_qza/smic_run4_
oct2023.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads_qza/s
mic_run4_oct2023.qzv &

nohup qiime demux summarize \
  --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads_qza/smic_run5_
apr2024.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads_qza/s
mic_run5_apr2024.qzv &

nohup qiime demux summarize \
  --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads_qza/smic_run6_
may2024.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads_qza/s
mic_run6_may2024.qzv &

nohup qiime demux summarize \
  --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads_qza/smic_run7_
may2024.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads_qza/s
mic_run7_may2024.qzv &

nohup qiime demux summarize \
  --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads_qza/smic_run8_
june2024.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads_qza/s
mic_run8_june2024.qzv &
```

Trim primers

```
nohup qiime cutadapt trim-paired \
  --i-demultiplexed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/r
eads_qza/smic_run4_oct2023.qza \
  --p-cores 6 \
  --p-front-f GTGYCAGCMGCCGCGGTAA \
  --p-front-r GGACTACNVGGGTWTCTAAT \
  --p-discard-untrimmed \
  --p-no-indels \
  --o-trimmed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads_q
za/smic_run4_oct2023-trimmed.qza &

nohup qiime cutadapt trim-paired \
  --i-demultiplexed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/r
eads_qza/smic_run5_apr2024.qza \
  --p-cores 6 \
  --p-front-f GTGYCAGCMGCCGCGGTAA \
```

```
   --p-front-r GGACTACNVGGGTWTCTAAT \
   --p-discard-untrimmed \
   --p-no-indels \
   --o-trimmed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads_q
za/smic_run5_apr2024-trimmed.qza &

nohup qiime cutadapt trim-paired \
   --i-demultiplexed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/r
eads_qza/smic_run6_may2024.qza \
   --p-cores 6 \
   --p-front-f GTGYCAGCMGCCGCGGTAA \
   --p-front-r GGACTACNVGGGTWTCTAAT \
   --p-discard-untrimmed \
   --p-no-indels \
   --o-trimmed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads_q
za/smic_run6_may2024-trimmed.qza &

nohup qiime cutadapt trim-paired \
   --i-demultiplexed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/r
eads_qza/smic_run7_may2024.qza \
   --p-cores 6 \
   --p-front-f GTGYCAGCMGCCGCGGTAA \
   --p-front-r GGACTACNVGGGTWTCTAAT \
   --p-discard-untrimmed \
   --p-no-indels \
   --o-trimmed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads_q
za/smic_run7_may2024-trimmed.qza &

nohup qiime cutadapt trim-paired \
   --i-demultiplexed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/r
eads_qza/smic_run8_june2024.qza \
   --p-cores 6 \
   --p-front-f GTGYCAGCMGCCGCGGTAA \
   --p-front-r GGACTACNVGGGTWTCTAAT \
   --p-discard-untrimmed \
   --p-no-indels \
   --o-trimmed-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads_q
za/smic_run8_june2024-trimmed.qza &
```

Review run specs after trimming

```
nohup qiime demux summarize \
   --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads_qza/smic_run4_
oct2023-trimmed.qza \
   --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads_qza/s
mic_run4_oct2023-trimmed.qzv &

nohup qiime demux summarize \
   --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads_qza/smic_run5_
```

```
apr2024-trimmed.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads_qza/s
mic_run5_apr2024-trimmed.qzv &

nohup qiime demux summarize \
  --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads_qza/smic_run6_
may2024-trimmed.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads_qza/s
mic_run6_may2024-trimmed.qzv &

nohup qiime demux summarize \
  --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads_qza/smic_run7_
may2024-trimmed.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads_qza/s
mic_run7_may2024-trimmed.qzv &

nohup qiime demux summarize \
  --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads_qza/smic_run8_
june2024-trimmed.qza \
  --o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads_qza/s
mic_run8_june2024-trimmed.qzv &
```

Review read quality summary in QIIME2 view

Run dada2 (based on how the read quality looks)

```
nohup qiime dada2 denoise-paired \
  --i-demultiplexed-seqs /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/reads_
qza/smic_run4_oct2023-trimmed.qza \
  --p-trunc-len-f 230 \
  --p-trunc-len-r 190 \
  --p-max-ee-f 4 \
  --p-max-ee-r 6 \
  --p-n-threads 8 \
  --output-dir /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output &

nohup qiime dada2 denoise-paired \
  --i-demultiplexed-seqs /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/reads_
qza/smic_run5_apr2024-trimmed.qza \
  --p-trunc-len-f 230 \
  --p-trunc-len-r 190 \
  --p-max-ee-f 4 \
  --p-max-ee-r 6 \
  --p-n-threads 8 \
  --output-dir /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output &

nohup qiime dada2 denoise-paired \
--i-demultiplexed-seqs /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/reads_qz
a/smic_run6_may2024-trimmed.qza \
```

```
--p-trunc-len-f 230 \
--p-trunc-len-r 190 \
--p-max-ee-f 4 \
--p-max-ee-r 6 \
--p-n-threads 8 \
--output-dir /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output &

nohup qiime dada2 denoise-paired \
--i-demultiplexed-seqs /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/reads_qz
a/smic_run7_may2024-trimmed.qza \
--p-trunc-len-f 230 \
--p-trunc-len-r 190 \
--p-max-ee-f 4 \
--p-max-ee-r 6 \
--p-n-threads 8 \
--output-dir /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output &

nohup qiime dada2 denoise-paired \
--i-demultiplexed-seqs /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/reads_qz
a/smic_run8_june2024-trimmed.qza \
--p-trunc-len-f 230 \
--p-trunc-len-r 190 \
--p-max-ee-f 4 \
--p-max-ee-r 6 \
--p-n-threads 8 \
--output-dir /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output &
```

Export dada2 stats and review

```
qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct202
3/dada2_output/denoising_stats.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/
smic_run4_oct2023/dada2_output

qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr202
4/dada2_output/denoising_stats.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/
smic_run5_apr2024/dada2_output

qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may202
4/dada2_output/denoising_stats.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/
smic_run6_may2024/dada2_output

qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may202
4/dada2_output/denoising_stats.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/
smic_run7_may2024/dada2_output/

qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun202
4/dada2_output/denoising_stats.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/
smic_run8_jun2024/dada2_output/
```

Check out the representative sequences and ASV tables in QIIME2 view

Representative sequences

```
qiime feature-table tabulate-seqs \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/representative_sequences.qza \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/representative_sequences-run4.qzv

qiime feature-table tabulate-seqs \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/representative_sequences.qza \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/representative_sequences-run5.qzv

qiime feature-table tabulate-seqs \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/representative_sequences.qza \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/representative_sequences-run6.qzv

qiime feature-table tabulate-seqs \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/representative_sequences.qza \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/representative_sequences-run7.qzv

qiime feature-table tabulate-seqs \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/representative_sequences.qza \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/representative_sequences-run8.qzv
```

ASV Table

```
qiime feature-table summarize \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/table.qza \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/table_summary.qzv \
--m-sample-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/smic_run4_oct2023-metadata.tsv

#average: 35,030.5


qiime feature-table summarize \
```

```
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/table.qz
a \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/
table_summary.qzv \
--m-sample-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/smic_r
un5_apr2024-metadata.tsv

#average: 39,299.5

qiime feature-table summarize \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/table.qz
a \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/
table_summary.qzv \
--m-sample-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/smic_r
un6_may2024-metadata.tsv

#average: 29,560.9

qiime feature-table summarize \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/table.qz
a \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/
table_summary.qzv \
--m-sample-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/smic_r
un7_may2024-metadata.tsv

#average: 54,281.6

qiime feature-table summarize \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/table.qz
a \
--o-visualization /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/
table_summary.qzv \
--m-sample-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/smic_r
un8_june2024-metadata.tsv

#average: 52,584.9
```

Filter out rare ASVs [do this per run, before merging]

```
qiime feature-table filter-features --i-table /mnt/Genomics/Working/ckellogg/sea-star/smi
c_run4_oct2023/dada2_output/table.qza \
--p-min-frequency 35 \
--p-min-samples 2 \
--o-filtered-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_outpu
t/table_filt_smic_run4.qza

qiime feature-table filter-features --i-table /mnt/Genomics/Working/ckellogg/sea-star/smi
```

```
c_run5_apr2024/dada2_output/table.qza \
--p-min-frequency 39 \
--p-min-samples 2 \
--o-filtered-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_outpu
t/table_filt_smic_run5.qza

qiime feature-table filter-features --i-table /mnt/Genomics/Working/ckellogg/sea-star/smi
c_run6_may2024/dada2_output/table.qza \
--p-min-frequency 29 \
--p-min-samples 2 \
--o-filtered-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_outpu
t/table_filt_smic_run6.qza

qiime feature-table filter-features --i-table /mnt/Genomics/Working/ckellogg/sea-star/smi
c_run7_may2024/dada2_output/table.qza \
--p-min-frequency 54 \
--p-min-samples 2 \
--o-filtered-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_outpu
t/table_filt_smic_run7.qza


qiime feature-table filter-features --i-table /mnt/Genomics/Working/ckellogg/sea-star/smi
c_run8_jun2024/dada2_output/table.qza \
--p-min-frequency 52 \
--p-min-samples 2 \
--o-filtered-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_outpu
t/table_filt_smic_run8.qza
```

Remove ASVs that were filtered out.

```
qiime feature-table filter-seqs --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run
4_oct2023/dada2_output/representative_sequences.qza \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/table_fi
lt_smic_run4.qza \
--o-filtered-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/
repseqs_filt_run4.qza

qiime feature-table filter-seqs --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run
5_apr2024/dada2_output/representative_sequences.qza \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/table_fi
lt_smic_run5.qza \
--o-filtered-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/
repseqs_filt_run5.qza

qiime feature-table filter-seqs --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run
6_may2024/dada2_output/representative_sequences.qza \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/table_fi
lt_smic_run6.qza \
--o-filtered-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/
```

```
repseqs_filt_run6.qza

qiime feature-table filter-seqs --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run
7_may2024/dada2_output/representative_sequences.qza \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/table_fi
lt_smic_run7.qza \
--o-filtered-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/
repseqs_filt_run7.qza

qiime feature-table filter-seqs --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run
8_jun2024/dada2_output/representative_sequences.qza \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/table_fi
lt_smic_run8.qza \
--o-filtered-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/
repseqs_filt_run8.qza
```

Update sample IDs before merging to avoid overlap

```
qiime feature-table rename-ids \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/table_fi
lt_smic_run4.qza \
--m-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/smic_run4_oct
2023-metadata.tsv \
--m-metadata-column 'SampleID_rev' \
--o-renamed-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/
table_filt_smic_run4_renamed.qza


qiime feature-table rename-ids \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/table_fi
lt_smic_run5.qza \
--m-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/smic_run5_apr
2024-metadata.tsv \
--m-metadata-column 'SampleID_rev' \
--o-renamed-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/
table_filt_smic_run5_renamed.qza

qiime feature-table rename-ids \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/table_fi
lt_smic_run6.qza \
--m-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/smic_run6_may
2024-metadata.tsv \
--m-metadata-column 'SampleID_rev' \
--o-renamed-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/
table_filt_smic_run6_renamed.qza

qiime feature-table rename-ids \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/table_fi
lt_smic_run7.qza \
```

```
--m-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/smic_run7_may
2024-metadata.tsv \
--m-metadata-column 'SampleID_rev' \
--o-renamed-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/
table_filt_smic_run7_renamed.qza

qiime feature-table rename-ids \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/table_fi
lt_smic_run8.qza \
--m-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/smic_run8_jun
e2024-metadata.tsv \
--m-metadata-column 'SampleID_rev' \
--o-renamed-table /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/
table_filt_smic_run8_renamed.qza
```

Merge Runs

```
qiime feature-table merge \
--i-tables /mnt/Genomics/Working/ckellogg/sea-star/from-fucus/16SV4_smic_merged/smic_runs
123_mergedtable.qza \
--i-tables /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/table_f
ilt_smic_run4_renamed.qza \
--i-tables /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/table_f
ilt_smic_run5_renamed.qza \
--i-tables /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/table_f
ilt_smic_run6_renamed.qza \
--i-tables /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/table_f
ilt_smic_run7_renamed.qza \
--i-tables /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/table_f
ilt_smic_run8_renamed.qza \
--o-merged-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs123456
78_mergedtable.qza
```

Merge representative sequences

```
qiime feature-table merge-seqs \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/from-fucus/16SV4_smic_merged/smic_runs12
3_merged_repseqs.qza \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run4_oct2023/dada2_output/repseqs_f
ilt_run4.qza \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run5_apr2024/dada2_output/repseqs_f
ilt_run5.qza \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run6_may2024/dada2_output/repseqs_f
ilt_run6.qza \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run7_may2024/dada2_output/repseqs_f
ilt_run7.qza \
--i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_run8_jun2024/dada2_output/repseqs_f
```

```
ilt_run8.qza \
--o-merged-data /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_mergedseqs.qza
```

Classify reads

```
#nb-classifier w/ silva

nohup qiime feature-classifier classify-sklearn \
--i-reads /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_mergedseqs.qza \
--i-classifier /mnt/Genomics/Working/ckellogg/db/dbs_qiime2022/silva/silva-138-99-515-806-nb-classifier.qza \
--o-classification /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_silva_classification.qza \
--p-n-jobs 10 &
```

Filter chloroplasts and mitochondria from feature table

```
qiime taxa filter-table \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_mergedtable.qza \
--i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_silva_classification.qza \
--p-exclude Mitochondria,Chloroplast,Eukaryota \
--o-filtered-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_mergedtable_NCM.qza
```

Filter chloroplasts and mitochondria from rep seqs

```
qiime taxa filter-seqs --i-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_mergedseqs.qza \
--i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_silva_classification.qza \
--p-exclude Mitochondria,Chloroplast,Eukaryota \
--o-filtered-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_mergedseqs_NCM.qza
```

Annotate dataset with GreenGenes database

1. Map to Greengenes2

```
qiime greengenes2 non-v4-16s \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_merg
edtable_NCM.qza \
--i-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_
mergedseqs_NCM.qza \
--i-backbone /mnt/Genomics/Working/ckellogg/db/greengenes2/2022.10.backbone.v4.fna.qza \
--o-mapped-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/seastar.gg2.qza
\
--o-representatives /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/seastar.gg2.
fna.qza
```

2. Classify!

```
qiime greengenes2 taxonomy-from-table \
--i-reference-taxonomy /mnt/Genomics/Working/ckellogg/db/greengenes2/2022.10.taxonomy.as
v.nwk.qza \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/seastar.gg2.qza \
--o-classification /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/seastar.gg2.t
axonomy.qza

#or using nb-classfier

nohup qiime feature-classifier classify-sklearn \
--i-reads /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_merg
edseqs_NCM.qza \
--i-classifier /mnt/Genomics/Working/ckellogg/db/greengenes2/2022.10.backbone.full-lengt
h.nb.sklearn-1.4.2.qza \
--o-classification /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_NC
M_gg2_nbclassification.qza \
--p-n-jobs 10 &
```

Remove some samples that are not part of this dataset

ASV table

```
qiime feature-table filter-samples \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_merg
edtable_NCM.qza \
--m-metadata-file /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_merged_me
tadata.tsv \
--p-where 'NOT [Species]="Mussel"' \
--p-filter-empty-features \
--o-filtered-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smiconly_NCM_
filtered-table.qza
```

Representative sequences

```
qiime feature-table filter-seqs --i-data /mnt/Genomics/Working/ckellogg/sea-star/smic_mer
ged_runs/smic_runs12345678_mergedseqs_NCM.qza \
--i-table /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smiconly_NCM_filtered-
table.qza \
--o-filtered-data /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_repseqs_N
CM_seastars.qza
```

Export [study samples only, all runs]

ASV table

```
qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/
smiconly_NCM_filtered-table.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/smi
c_merged_runs/exported_runs1to8/
```

Representative sequences

```
qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/
smic_repseqs_NCM_seastars.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/smic_
merged_runs/exported_runs1to8/
```

Do some exporting [all samples all runs]

Representative sequences

```
qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/
smic_runs12345678_mergedseqs_NCM.qza --output-path /mnt/Genomics/Working/ckellogg/sea-sta
r/smic_merged_runs/exported_runs1to8/
```

ASV table

```
qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/
smic_runs12345678_mergedtable_NCM.qza --output-path /mnt/Genomics/Working/ckellogg/sea-st
```

```
ar/smic_merged_runs/exported_runs1to8/
```

Silva Taxonomy

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_ru
ns12345678_silva_classification.qza --output-path /mnt/Genomics/Working/ckellogg/sea-sta
r/smic_merged_runs/exported_runs1to8/
```

Green Genes Taxonomy

```
qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/
seastar.gg2.taxonomy.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merge
d_runs/exported_runs1to8/
```

Green Genes Taxonomy (nb classifier)

```
qiime tools export --input-path /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_ru
ns12345678_NCM_gg2_nbclassification.qza --output-path /mnt/Genomics/Working/ckellogg/sea-
star/smic_merged_runs/exported_runs1to8/
```

```
qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/
seastar.gg2.qza --output-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/ex
ported_runs1to8/
```

Grab the ASVs that are not classified beyond bacteria.

```
qiime taxa filter-seqs --i-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_
runs/smic_runs12345678_mergedseqs.qza \
--i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_silva_cl
assification.qza \
--p-exclude p__ \
--o-filtered-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs
12345678_mergedseqs_unclassified.qza
```

```
qiime tools export --input-path /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/
smic_runs12345678_mergedseqs_unclassified.qza --output-path /mnt/Genomics/Working/ckellog
```

```
g/sea-star/smic_merged_runs/exported_runs1to8/
```

```
qiime taxa filter-seqs --i-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_
runs/smic_runs12345678_mergedseqs.qza \
--i-taxonomy /mnt/Processed/ckellogg/sea-star/smic_merged_runs/smic_runs12345678_silva_cl
assification.qza \
--p-exclude p__ \
--o-filtered-sequences /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/smic_runs
12345678_mergedseqs_unclassified.qza
```

Make BIOM file

Change the taxonomy header so that it is as follows:

#OTUID taxonomy confidence

```
biom add-metadata -i /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/exported_ru
ns1to8/smic-all-runs1to8-NCM.biom -o /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_
runs/exported_runs1to8/smic-all-runs1to8-NCM-tax.biom --observation-metadata-fp /mnt/Geno
mics/Working/ckellogg/sea-star/smic_merged_runs/exported_runs1to8/smic-silva-taxonomy.tsv
--sc-separated taxonomy


biom add-metadata -i /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/exported_ru
ns1to8/smiconly-runs1to8-NCM.biom -o /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_
runs/exported_runs1to8/smiconly-runs1to8-NCM-tax.biom --observation-metadata-fp /mnt/Geno
mics/Working/ckellogg/sea-star/smic_merged_runs/exported_runs1to8/smic-silva-taxonomy.tsv
--sc-separated taxonomy

biom add-metadata -i /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/exported_ru
ns1to8/smiconly-runs1to8-NCM.biom -o /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_
runs/exported_runs1to8/smiconly-runs1to8-NCM-gg2tax.biom --observation-metadata-fp /mnt/G
enomics/Working/ckellogg/sea-star/smic_merged_runs/exported_runs1to8/smic-gg2-nb-taxonom
y.tsv --sc-separated taxonomy

biom add-metadata -i /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/exported_ru
ns1to8/smic-all-runs1to8-NCM.biom -o /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_
runs/exported_runs1to8/smic-all-runs1to8-NCM-gg2tax.biom --observation-metadata-fp /mnt/G
enomics/Working/ckellogg/sea-star/smic_merged_runs/exported_runs1to8/smic-gg2-nb-taxonom
y.tsv --sc-separated taxonomy
```

```
biom convert -i /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/exported_runs1to
8/smiconly-runs1to8-NCM-tax.biom -o /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_r
uns/exported_runs1to8/asvtable_NCM_smiconly-silvatax.tsv --to-tsv --header-key taxonomy
```

```
biom convert -i /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/exported_runs1to
8/smiconly-runs1to8-NCM-gg2tax.biom -o /mnt/Genomics/Working/ckellogg/sea-star/smic_merge
d_runs/exported_runs1to8/asvtable_NCM_smiconly-gg2tax.tsv --to-tsv --header-key taxonomy

biom convert -i /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/exported_runs1to
8/smic-all-runs1to8-NCM-gg2tax.biom -o /mnt/Genomics/Working/ckellogg/sea-star/smic_merge
d_runs/exported_runs1to8/smic-all-runs1to8-NCM-gg2tax.tsv --to-tsv --header-key taxonomy

biom convert -i /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_runs/exported_runs1to
8/smic-all-runs1to8-NCM-tax.biom -o /mnt/Genomics/Working/ckellogg/sea-star/smic_merged_r
uns/exported_runs1to8/smic-all-runs1to8-NCM-silvatax.tsv --to-tsv --header-key taxonomy
```